
PRACTICE EXAM 2

NAME: _____

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Consider the matrix $A = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$. Solve the following questions concerning A :

(a) Find A^{-1} .

(b) Express A as a product of elementary matrices.

Problem 2. Answer the following questions regarding linear transformations.

(a) State precisely what it means for a transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be linear.

(b) Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the matrix transformation induced by the $m \times n$ matrix A . Show that T is a linear transformation.

- (c) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects every vector $\mathbf{x}$ along the x -axis, and then reflects the result along the y -axis. Find the matrix of T and compute the image of $(1, 2)$ under T .
(**Hint.** Draw some pictures to understand the effect of the transformation T)

Problem 3. (a) Using determinants, find the values of c for which the matrix

$$A = \begin{bmatrix} c & 1 & 1 \\ 1 & -c & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

is **not** invertible.

(b) Calculate the determinant of

$$B = \begin{bmatrix} -3 & 0 & 1 & -3 & -2 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 3 & 1 \\ 0 & 0 & 7 & 2 & 0 \end{bmatrix}.$$

Problem 4. On each of the following questions, **completely fill in** the circle next to the correct answer. Each question only has one correct answer. You do not need to justify your choices.

- (a) If A is a 3×3 matrix and $E_{R_1-2R_3}$ is the 3×3 elementary matrix associated with the row operation $R_1 - 2R_3$, then the result of applying the row operation $R_1 - 2R_3$ to A is:

- ☐ $AE_{R_1-2R_3}$.
- ☐ $E_{R_1-2R_3}A$.
- ☐ $E_{R_1-2R_3}^{-1}A$.
- ☐ $AE_{R_1-2R_3}^{-1}$.

- (b) Coming back to the matrix $E_{R_1-2R_3}$, its determinant is equal to

- ☐ $\det(E_{R_1-2R_3}) = 1$.
- ☐ $\det(E_{R_1-2R_3}) = -1$.
- ☐ $\det(E_{R_1-2R_3}) = 2$.
- ☐ $\det(E_{R_1-2R_3}) = -2$.

- (c) If A and B are $n \times n$ invertible matrices, then

- ☐ AB^T is invertible and its inverse is $A^{-1}(B^{-1})^T$.
- ☐ AB^T is invertible and its inverse is $(B^{-1})^T A^{-1}$.
- ☐ AB^T is invertible and its inverse is BA^T .
- ☐ AB^T is not invertible.

- (d) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = 3\sqrt{x} + y^3$ is linear.

- ☐ True.
- ☐ False.

- (e) If A is an $n \times n$ matrix, which of these conditions implies that A is invertible?

- ☐ The system $A\mathbf{x} = \mathbf{0}$ has two different solutions.
- ☐ $A^3 = I_n$.
- ☐ There exists a nonzero matrix B such that $AB = 0$.
- ☐ A has two equal rows.

Scratch working page